

Fact Sheet

CHINESE MILITARY AND INTELLIGENCE RENDEZVOUS AND PROXIMITY OPERATIONS

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SUMMARY

Since 2008, China has conducted multiple tests of satellite technologies for robotic rendezvous and proximity operations (RPO) to support surveillance and inspection of other space objects in both low Earth orbit (LEO) and geostationary Earth orbit (GEO), most of which are related to military or intelligence operations. While these missions have been publicly acknowledged by China, most have few public details and a few remain shrouded in secrecy. None of the programs listed here have strong evidence to link them to co-orbital anti-satellite (ASAT) testing or deployment.

RENDEZVOUS & PROXIMITY OPERATIONS

Proximity operations are a series of orbital maneuvers executed to place and maintain a spacecraft in the

vicinity of another space object on a relative planned path for a specific time duration to accomplish mission objectives. Rendezvous is a process whereby two space objects, artificial or natural bodies, are intentionally brought close together through a series of orbital maneuvers at a planned time and place. Taken together, RPO technologies enable a wide range of capabilities to support civil and commercial space activities such as on-orbit inspections, repair, refueling, assembly, and life extension. RPO capabilities can also be used for military and intelligence space activities such as intelligence, surveillance, and offensive weapons such as co-orbital anti-satellites.

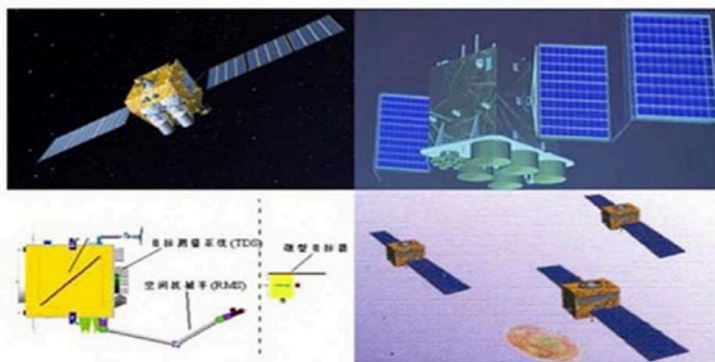
CHINESE MILITARY AND INTELLIGENCE RPO MISSIONS IN LEO

Since the late 2000s, China has conducted a series of robotic on-orbit demonstrations between different sets of satellites. The first known Chinese robotic RPO occurred in September 2008 when a Chinese human spaceflight mission Shenzhou 7 deployed a small satellite to practice on-orbit inspection and control system flying capabilities.¹ Some observers concluded that the BX-1 was a test of the capabilities

required for a co-orbital ASAT attack² but the mission has not been linked to any such military program.³

In the summer of 2010, the Chinese satellites Shi Jian (SJ)-12 and SJ-06F conducted a series of robotic RPO.⁴ The mission of SJ-12, as stated by the state media service *Xinhua*, is to carry out “scientific and technological experiments.”⁵ In the summer of 2010,

附图 1: 2013 年前三季度公司小卫星产品发射交付概览



资料来源: 上海证券研究所

Image of the SY-7 (lower left, with robotic arm) and its small companion satellite.

Image credit: Liss

the SJ-12 initiated a series of deliberate changes in its orbital trajectory to approach and rendezvous with the SJ-06F satellite.⁶ The maneuvers occurred over several weeks between June 12, 2010, and August 16, 2010. On August 19, the two satellites came within 300 meters. A change in the orbital trajectory for the SJ-06F around that same time indicates that the two satellites may have bumped into each other, although there were no external indications of damage to either satellite or any debris created by the incident.

On July 19, 2013, China placed three payloads into roughly similar LEO orbits from the same launch: Shiyao 7 (SY-7), Chuangxin 3 (CX-3), and Shijian 15 (SJ-15).⁷ The SY-7 carried a teleoperated robotic arm while the SJ-15 was equipped with RPO capabilities. In August 2013, SJ-15 conducted an RPO of CX-3 within a few kilometers and another RPO with Shijian 7 (SJ-7), a Chinese satellite launched in 2005.⁸ Anonymous US officials claimed that the RPO was part of a “covert anti-satellite weapons development program,” and that one of the satellites “grabbed” another,⁹ although the satellite with the arm, SY-7, was not involved. On October 18, 2013, the SY-7 initiated a small maneuver to raise its orbit by several hundred meters and shortly thereafter released another object, Payload A, which orbited in relatively close proximity for several days. Some reports claimed the two objects may have physically joined with each other,¹⁰ but the publicly available tracking was not accurate enough to confirm. Both objects occasionally conducted small maneuvers throughout 2014 and 2015, and the SJ-15 through 2016 as well.¹¹

In April 2014, the SJ-15 began another series of small maneuvers to conduct proximity operations around the CX-3. Between April 12-14, the SJ-15 raised its orbit by several tens of kilometers, and then between May 12 and 14, Payload C lowered its orbit by several tens of kilometers. The effect of these maneuvers was to match orbital planes once again with the SJ-7, and on a trajectory that brought it above and then behind the SJ-7 at a range of around 150 km, with a vertical separation of a few kilometers.¹² Throughout the rest of May, the SJ-15 slowly decreased the distance to the SJ-7 to within a kilometer.¹³ The SJ-15 continued to occasionally make changes to its orbit in 2015 and 2016, but the reasons for doing so were unclear. On May 6, 2016, the SJ-15 changed its altitude by several tens of kilometers, bringing it close to the CX-3 again.¹⁴

In 2016, another Chinese satellite was launched that again created concerns about on-orbit grappling. The Aolong-1, also known as the Advanced Debris Removal Vehicle (ADRV) or “Roaming Dragon,” was a small satellite developed by Harbin Institute of Technology under contract to the China Academy of Launch Vehicle Technology (CALT) to reportedly demonstrate using a robotic arm to capture a small piece of space debris for removal from orbit. Aolong-1 was placed into orbit on the first launch of China’s new Long March 7 (LM-7) rocket on June 25, 2016, from Wenchang Space Launch Center, along with a scaled-down test version of China’s next human spacecraft, a ballast mass, and a few small rideshare cubesats. The purpose of the launch was to demonstrate the ability of the LM-7 and its restartable upper stage to place the new crewed spacecraft into orbit, to deploy multiple payloads into different orbits, test the new Tianyuan-1 refueling system developed by the National University of Defense Technology, and test the atmospheric re-entry of the crewed spacecraft test vehicle.¹⁵ Although they were only small parts of the mission, the debris removal and refueling experiments generated significant press outside of China due to concerns over dual-use technology and China leaping ahead in technology. The reality of either the Aolong-1 or the refueling experiment was less than the media hype. By all appearances, the Tianyuan-1 refueling system was attached to the upper stage, as no separate satellite of that description was ever catalogued by the US military, nor did any of the ten objects catalogued in space rendezvous with any other satellites. According to US military tracking data, the Aolong-1 did indeed separate into a 380 km by 200 km orbit but did not rendezvous with any other objects. The debris capture experiment appears to have been simulated, and the Aolong-1 does not appear to have altered its orbit during its short two months on orbit.¹⁶

In April 2021, China launched nine small payloads into LEO from Taiyuan Space Launch Center that reportedly carried an orbital debris removal experiment.¹⁷ The NEO-01 payload, developed by Origin Space Technology Company, reportedly carried out an experiment of using a large net to capture a tethered target simulating orbital debris.¹⁸ However, it is difficult to accurately verify if this claim is true as it is unclear if any of the objects from the launch released any additional objects. None appeared to have approached any existing pieces of space debris.

SJ-6 05A (SJ-6I) and SJ-6 05B (SJ-6J) were launched in December 2021 via a Long March 4B from the Jiuquan Satellite Launch Center into a SSO.¹⁹ CASC is reported to have said that their mission was for space environment exploration and technology verification.²⁰ The pair was initially at an altitude of 470 km and conducted RPOs in 2022, when SJ-6 05A got within 1 km of SJ-6 05B repeatedly starting that March.²¹ In October 2022, they completed their RPOs and raised their orbits about 100 km to join the rest of the SJ-6 series.²²

CASI's Kristin Burke published in July 2025 an analysis of the SJ-6 series, at which point consisted of five pairs.²³ She identified three consistencies across the series: "The A satellite maneuvers more than the B satellite. The B satellite acts as a target for another satellite's maneuvers, either in series or out of series. The SJ-6 pairs contribute to RPO TTPs for a satellite outside of their series. For example, SJ-12, and more recently SY-24C."²⁴ She noted that this most recent series of RPOs had "important consistencies with past behavior and is an incremental advancement for improved payload testing and TTP [tactics, techniques, and procedures] development," and suggested that the "A" satellites in future SJ-6 series would continue to get more maneuverable.²⁵ Finally, she urged analysts to look back at the whole series to identify new capabilities versus expanded ones, because "Without identifying past similarities, one is more likely to interpret behavior as a near term threat, rather than an evolving and manageable challenge."²⁶

On December 25, 2023, a Long March 11 (CZ-11) launched three satellites from a barge launch pad near Guangdong.²⁷ The three satellites, Shiyan-24C satellites (SY-24C 01, SY-24C 02, SY-24C 03) - were inserted into a Sun-synchronous orbit inclined at 97.3 degrees (at an altitude of 540 x 553 x 540 km). Their orbit was co-planar with SJ-6 05A and SJ-6 05B, two satellites launched on October 12, 2021, via a Long March 4B (CZ 4B) launch vehicle from the Jiuquan Satellite Launch Center.²⁸ According to a USSF fact sheet, the five satellites started conducting RPOs from mid-March 2024 through the end of April 2024, at times separated by less than 1km and undertaking "two simultaneous proximity events at the same time."²⁹ In September 2024, SY-24C 02 and SY-24C 03 conducted three separate approaches, again under 1km, with two being multi-day actions.³⁰ The following December, in another multi-day event, SY-24C 03 and SJ6 5A came within "tens of meters" of each other five different times and with "associated relative velocities less than 10 cm/s."³¹ After that, the five satellites maneuvered to maintain a separation of over 100 km.³²

CHINESE MILITARY AND INTELLIGENCE RPO MISSIONS IN GEO

China has also conducted robotic RPO demonstrations in GEO. On November 3, 2016, China placed the SJ-17 satellite in GEO, which was publicly declared to be designed to test advanced technologies.³³ Several days after reaching GEO, the SJ-17 began maneuvering to place itself into the active GEO belt close to another Chinese satellite, Chinasat 5A, at a longitude of 162.9 E.³⁴ The SJ-17 made several small maneuvers to circumnavigate Chinasat 5A at a distance of between 50 to 100 km for several days, slowly closing in to within a few km on November 30, before returning to a 50 to 100 km standoff distance.³⁵ The two satellites remained close until December 29, when Analytical Graphics, Inc, (AGI) reported that Chinasat 5A was drifting away.³⁶

The SJ-17 drifted eastward then sharply westward until March 2018, when it lowered its orbit to reverse its drift. This indicated it conducted a survey of the GEO region. Over the first half of 2018, the SJ-17 made further unusual changes to its orbit. SJ-17's reversal in inclination and maneuvering to a drift orbit between January and July of 2018 appears to be linked to an unexplained anomaly in the orbital trajectory of Chinasat 1C, a Chinese communications satellite launched in December 2015.³⁷ This sudden, large change in inclination suggests the SJ-17 possesses a significant delta-v capability. SJ-17 rendezvoused with Chinasat 1C, coming within 1.5 km on July 29. Ten days later, Chinasat 1C halted its drift and slowly drifted back to its operational location. SJ-17 remained with Chinasat 1C until August prior to departing, while Chinasat 1C arrived back at its original location on September 7. This strongly suggests that SJ-17 inspected Chinasat 1C to determine the source of the anomaly and monitor the recovery attempt.

Additionally, in January and October 2020, SJ-17 made smaller changes to RPO with Chinasat 6B and SJ-20. SJ-17 started drifting eastward in February 2023; in November 2023, SJ-17 spent a brief period next to VENESAT-1, a Venezuelan communications satellite built by China and launched in 2008. SJ-17 continued to move, changing its altitude six times in the process (five times taking it below the GEO belt); it dropped its altitude by 76.2 km to move below the GEO belt on Nov. 1, 2024 and then increased its altitude by 76.2 km to rejoin the GEO belt on Nov. 14.³⁸ It officially entered the Western hemisphere on Nov. 20, 2024 and appears to be stable at 178 degrees West longitude, with the nearest operational satellite being Russia's YAMAL 300K.³⁹

On December 23, 2018, China launched another mission to GEO that also exhibited unusual behavior. Like its predecessors, the Tongxin Jishu Shiyan (TJS)-3 satellite was launched from Xichang Space Launch Center into an elliptical geosynchronous transfer orbit (GTO). Chinese official media has described them as communications technology test satellites but observers believe they may also be testing missile warning sensors, deployable antennas, or other technology.⁴⁰ TJS-3 appeared to be similar in nature, and the US military ended up cataloging two objects from the launch in GEO: the TJS-3 satellite and a second object (object 43917) that was assumed to be an apogee kick motor (AKM), a detachable rocket engine often used to circularize a satellite in GEO, as it was slowly drifting westward. However, shortly after the separation, object 43917 did a series of maneuvers to place it into a GEO slot at 59.07E, near TJS-3.⁴¹ Object 43917 slowly drifted toward TJS-3 and, according to AGI, exhibited photometry consistent with a stabilized object, indicating it was a functional satellite and not an AKM.⁴²

Since May 2019, TJS-3 has circled the GEO belt and remained relatively close (within a couple of hundred kilometers) to multiple satellites, including USA 263 in July 2019 and USA 233 in March 2021 and October 2022. Given that distance, this activity is not technically a RPO but is still likely to generate concerns. TJS-3 moved into 173.5 degrees east longitude in May 2023. Another Chinese satellite, TJS-10, was launched on November 3, 2023, on a Long March 7A from Wenchang Satellite Launch Center; Chinese state media said it was intended to be "mainly used for multi-band and high-speed communication technology experiments."⁴³ By November 9, it had circularized its orbit and settled into GEO at 173.3 degrees east longitude. TJS-10 and TJS-3 spent about a year undertaking coordinated maneuvers in close proximity to each other (under 50 km) as they kept synchronized orbits with very similar inclination.⁴⁴ In November 2024, TJS-3 separated from TJS-10 in order to move around the GEO belt.⁴⁵ TJS-3 continued to move until reaching 148.7 degrees East in January 2025.⁴⁶ The satellite subsequently reduced altitude by 53 km to stop its westward drift and was placed back on the GEO belt at 148.23 degrees East with minimal drift.⁴⁷

On October 24, 2021, China launched a classified satellite from the Xichang satellite launch center that it claimed was for a space debris mitigation mission.⁴⁸ The satellite, publicly named SJ-21, was placed into an initial GTO. By November 2, the SJ-21 used an apogee kick motor to circularize its orbit at about 156E and bring the inclination down to 8°, releasing the AKM as a piece of debris afterward. For a while, SJ-21 maintained close proximity to the AKM, which suggested it was conducting an RPO.⁴⁹

On December 25, 2021, the SJ-21 rendezvoused with a defunct Chinese navigation satellite, Compass G2. The Compass G2 was a second-generation navigation satellite launched in 2009 as part of China's Beidou constellation and appeared to fail early in its orbital lifetime as it lost station keeping and began to drift both east-west and increase in inclination. Compass G2 also experienced a fragmentation event in 2016 that released at least six trackable pieces of debris.⁵⁰ While maintaining tight proximity to Compass G2 for several weeks, the SJ-21 docked to it at some point and then around January 21, 2022, used its onboard propulsion to pull both satellites to a higher altitude above the geostationary belt. By January 27, 2022, both objects were in an elliptical orbit ranging from 290 km to 3,100 km above the protected GEO zone, as observed by commercial trackers.⁵¹ Shortly thereafter, SJ-21 reduced its orbital altitude back down to close to GEO.

On December 23, 2021, China launched a pair of satellites into GEO orbit as part of the Shiyang series, officially labeled Shiyang (SY)-12 01 and Shiyang (SY)-12 02.⁵² The two satellites remained relatively close to each other in GEO, indicating that they had maneuvering capability and may have been conducting RPO. In late January 2022, USA 270 maneuvered to approach SY-12 01 and SY-12 02. According to tracking data collected by ExoAnalytic Solutions, SY-12 01 and SY-12 02 made significant maneuvers to split up and begin rotating around the GEO belt in opposite directions, with SY-12 02 apparently also getting an imaging opportunity on USA 270.⁵³ In September 2025, SY-12 02 conducted an RPO getting within 60 km of USA 336, the United States' SBIRS GEO 6 spacecraft.⁵⁴ Greg Gillinger noted, "The solar phase angle was ideal for SY-12 02 to image USA 336 with the sun nearly directly behind SY-12 02" but that it is "[u]nknown if China conducted this maneuver to optimize collection opportunity with USA 336."⁵⁵

On January 8, 2023, China launched a satellite identified as SJ-23 into a GTO. China state media described the satellite as being used for "scientific experiments and technical verification" but provided no further details.⁵⁶ Around January 15, the SJ-23 appeared to release another object, which was cataloged as an apogee kick motor but may be a functioning satellite.⁵⁷ Analysis of the orbital data suggests that the two objects got within 10 km of each other.

Shijian (SJ)-25 was launched into what appears to be a GTO on January 6, 2025, via a Long March 3B rocket from Xichang Satellite Launch Center.⁵⁸ It was made and built by SAST, who said it was intended to be used for "satellite fuel replenishment and life extension service technology verification."⁵⁹ In early June 2025, SJ-21 was detected maneuvering by commercial SSA companies; it was drifting westward along the GEO belt at 3.96 degrees/day on a trajectory to SJ-25.⁶⁰ Two US GSSAP satellites, USA 270 and USA 271, maneuvered along the GEO belt as well, positioning themselves so that they were "flanking" SJ-21 and SJ-25, in the words of US commercial SSA company COMSPOC, most likely to monitor the two Chinese satellites' activities.⁶¹ The two satellites were seen conducting RPOs June 13-14 and at times were unable to be distinguished from each other.⁶²

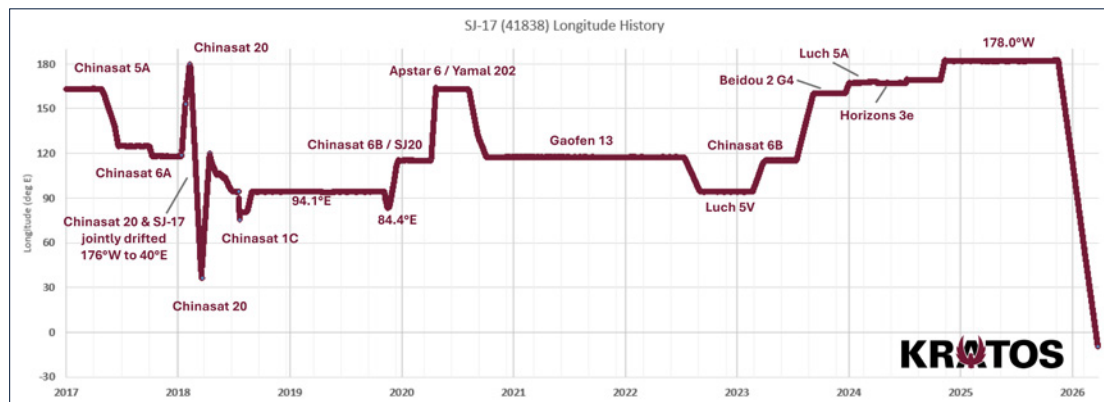
According to COMSPOC, on June 13, 2025, the two satellites got within 1 km of each, possibly docked, and then separated.⁶³ They again appeared to come close enough to dock on June 30 according to imagery from Swiss commercial SSA company s2a systems.⁶⁴ They got so close to each other between July 2 and July 6 that they were thought to have docked.⁶⁵ The SJ-21 and SJ-25 remained docked and started conducting minor maneuvers the first week of August; the second week of August, they conducted

energy intensive, plane-changing maneuvers that reduced their inclination by 6°. ⁶⁶ One analyst noted that this maneuver used the equivalent of six years of regular station-keeping fuel. ⁶⁷ By August 16, they had identical orbital information, were orbiting 3 km below the GEO belt at 127°E and a 4.4° inclination. ⁶⁸ It is not clear why they moved, the Chinese government has not released any information about them, but one analyst pointed out that they were now in a similar orbit plane as classified objects from USSF-44 launch in November 2022. ⁶⁹

s2a systems released images dated November 29, 2025, that showed the two satellites had undocked. ⁷⁰ This was further confirmed by COMSPOC who said that SJ-25 had conducted a burn on November 25 “consistent with an undock or separation maneuver” and that while the two satellites were 10 km apart, they were distinguishable from each other. ⁷¹ Analysts believe that SJ-25 had served as a gas station for SJ-21. ⁷²

SJ-21 and SJ-25 maneuvered mid-December to get within 1.3 km of each other on December 18; by December 20, optical imaging could not distinguish the two from each other again; and then by December 22, the two objects were distinct from one another. ⁷³ The two satellites conducted RPOs in the first half of January 2026, often staying within 5 km of each other and having their closest approach, just under 3 km, on January 13; by January 16, they were 130 km apart and SJ-25 was moving away from SJ-21 at a rate of 50 km/day (while still remaining co-planar). ⁷⁴

Research emerged that since 2018, SAST advertised “orbital mission extension vehicles” that could refuel satellites in GEO. ⁷⁵ It advertised what it called an “on-orbit gas station” at the 2018 Zhuhai Air Show. ⁷⁶ The customer satellite would need SAST’s refuelling docking port and the refueler would use two robotic arms to connect once it got within two meters. ⁷⁷

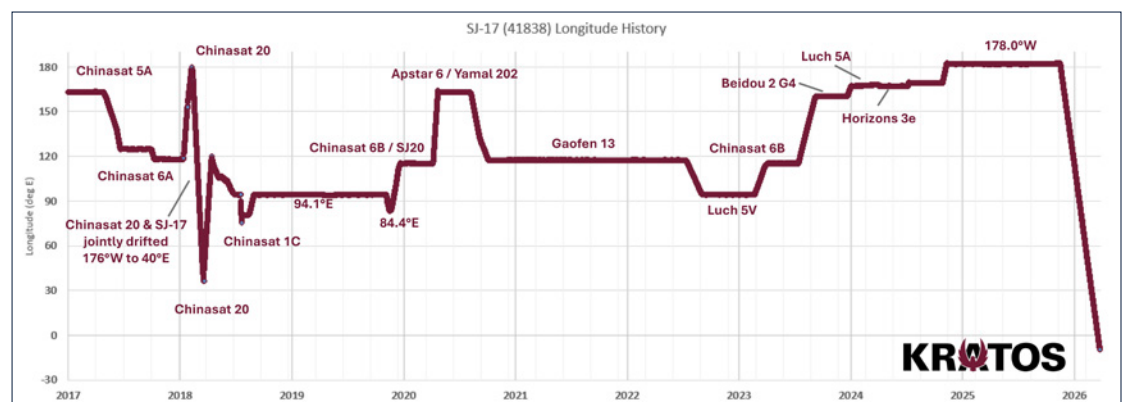


The longitudinal history of the SJ-17 satellite, demonstrating major RPOs with other satellites.

Image credit: Kratos

The longitudinal history of the TJS-3 satellite demonstrating major RPOs with other satellites.

Image credit: Kratos



DUAL-USE CAPABILITIES

The activities of the aforementioned Chinese RPO satellites are consistent with the demonstration of RPO technologies for satellite servicing, space situational awareness, and inspection. The most likely military utility of the capabilities demonstrated by the SJ-12, SJ-15, SY-7, SJ-17, TJS-3 AKM, TJS-3, SJ-21, SY-12 01 and 02, and SJ-23 satellites is for on-orbit space situational awareness (SSA) and satellite servicing. Their operational pattern was consistent with slow, methodical, and careful approaches to rendezvous with other space objects in similar orbits. The satellites the SJ-12 and SJ-15 approached were in relatively similar orbits, differing in altitude by a couple of hundred kilometers and slightly in inclination. They did not make huge changes to rendezvous with satellites in significantly different orbits. This behavior is similar to several US RPO missions to test and demonstrate satellite inspection and servicing capabilities such as the XSS-11. Notably, a counterspace assessment released by the Defense Intelligence Agency (DIA) in February 2019 stated that China is developing capabilities for inspection, repair, and space debris removal that may also be used as a weapon but did not specifically state that any Chinese RPO activities were a weapons test.⁷⁸

The SJ-17's approach to Chinasat 5A was not inconsistent with the way other active satellites in the GEO belt relocate to different orbital slots. It is also not unusual for satellites to be co-located within several tens of kilometers to share a GEO slot, although it is rare for them to approach within 1 km as the SJ-17 eventually did. Such a close approach in GEO could be used for very detailed imaging or inspection of another satellite or to intercept radio frequency signals directed at another satellite from Earth. Likely examples of the latter are the activities of the US PAN satellite between 2009 and 2014, and the Russian Luch/Olymp satellite beginning in 2014.

That said, there is evidence to suggest there is more than one category of RPO missions being performed. For example, the SJ-17 has maintained long periods of time relatively close to other communications satellites, while the SY-12 01 and 02 pair have largely been drifting through the GEO belt. This suggests they are being used for a different purpose.

While the known on-orbit activities of the satellites did not include explicit testing of offensive capabilities or aggressive maneuvers, it is possible that the technologies they tested could be used for offensive purposes in the future. One potential offensive use would be to get a radio-frequency jammer close to a satellite, thereby greatly amplifying its ability to interfere with the satellite's communications. While possible, there is no direct public evidence of such systems being tested on orbit, although there have been multiple research articles published in Chinese journals discussing and evaluating the concept.⁷⁹

While some skepticism exists relating Chinese rendezvous and proximity activities with co-orbital ASAT testing, there is no strong publicly available evidence linking these missions and a defined defense program. However, at least one recent Chinese research paper suggests using RPO capabilities to plant small explosive charges in the nozzle of a spacecraft's engine.⁸⁰ To date, China has not conducted any confirmed co-orbital ASAT testing on orbit, only testing of ground-launched direct-ascent ASAT weapons.

The onboard tracking and guidance systems used for rendezvous could be used to try and physically collide with another satellite to damage or destroy it. However, the approach would have to involve much higher relative velocities than what the Chinese RPO satellites have demonstrated to date, and potentially involve higher velocities and longer closing distances than what these satellites are capable of. Furthermore, the deliberate maneuvering to create a conjunction with the target satellite would be detectable with existing processes already in place to detect accidental close approaches. The warning time of such a close approach would likely be at least hours (for LEO) or days (for GEO) unless the attacking satellite was already in a very similar orbit.

If SJ-25 did indeed refuel SJ-21, this capability would be of great utility, as China's inspector satellites could stay on orbit and continue their missions while having the flexibility of maneuvering as the operators wished. If SJ-25 is a refueling satellite, it is unclear how much extra fuel SJ-25 has.

Summary of Known or Suspected Chinese Rendezvous & Proximity Operations in Space

Date	Target Satellite	Chaser Satellite	Result
Sep. 2008	SZ-7	BX-1	BX-1 was deployed from SZ-7 and proceeded to orbit around the spacecraft, taking images.
Jun. - Aug. 2010	SJ-06F	SJ-12	SJ-12 moved to rendezvous with SJ-06F. Satellites may have bumped into each other.
Jul. 2013 - May 2016	SJ-7, CX-3	SY-7, Payload A Debris, CX-3, SJ-15	SY-7 released an additional object (Payload A debris) that it performed maneuvers with and may have had a robotic arm. CX-3 performed optical surveillance of other in-space objects. SJ-15 demonstrated altitude and inclination changes to approach other satellites. In Dec. 2015, SJ-15 raised its orbit to CX-3, demonstrating multi-orbit inspection.
Nov. 2016 - Aug. 2018	Chinasat 5A, Chinasat 20, Chinasat 1C	SJ-17	SJ-17 demonstrated maneuverability around the GEO belt and circumnavigated Chinasat 5A. The SJ-17 then lowered its orbit to reverse its drift and moved to RPO with Chinasat 20. It undertook a sudden, large change in inclination to rendezvous and follow Chinasat 1C for about a week, which had undergone an anomaly in its orbital trajectory and started drifting westward. This strongly suggests that SJ-17 was used to inspect Chinasat 1 to determine the source of the anomaly and then monitor the recovery attempt.
Jan. - Apr. 2019	TJS-3, TJS-3 AKM	TJS-3, TJS-3 AKM	TJS-3 AKM separated from the TJS-3 in the GEO belt, and both performed small maneuvers to maintain relatively close orbital slots. Both satellites then maneuvered away from each other.
May 2019 - Present	Luch, USA 233, USA 263, Chinasat 10, Chinasat 16, SJ-20, Chinasat 12, TJS-10	TJS-3	TJS-3 is drifting around the GEO belt, periodically stopping to conduct RPO with other satellites.
Jan. - Oct. 2020	Chinasat 6B, SJ-20	SJ-17	SJ-17 conducted RPO with Chinasat 6B in January 2020 and SJ-20 in October 2020.
Dec. 2021 - Jan. 2022	Compass G2	SJ-21	SJ-21 maneuvered to dock with Compass G2 and pull it well past graveyard orbit.
Jan. 2022	SY-12 (01), SY-12 (02)	USA-270	USA 270 maneuvered to approach SY-12 (01) and SY-12 (02). These two Chinese satellites made significant maneuvers to split up and begin rotating around the GEO belt in opposite directions, with SY-12 (02) apparently also getting an imaging opportunity on USA 270.
Mar. - Oct. 2022	SJ-6 05B	SJ-6 05A	SJ-6 05A maneuvered to get within 1 km of SJ-6 05B repeatedly.
Jan. - Feb. 2023	SJ-23 AKM	SJ-23	Shortly after its launch, SJ-23 appeared to release an object, which was cataloged as an apogee kick motor but may be a functioning satellite. Analysis of the orbital data suggests that the two objects got within 10 km of each other.

Date	Target Satellite	Chaser Satellite	Result
Feb. 2023 - present	VENESAT-1, YAMAL 300K	SJ-17	SJ-17 started drifting eastward in Feb. 2023 and made a brief RPO in Nov. 2023 with VENESAT-1. SJ-17 continued to move, changing its altitude repeatedly to move above and below the GEO belt; it rejoined the GEO belt in November 2024. It officially entered the Western hemisphere on Nov. 20, 2024 and appears to be stable at 178 degrees West longitude, with the nearest operational satellite being Russia's YAMAL 300K.
Mar. - Apr., Sep., Dec. 2024	SY-24C 01, SY-24C 02, SY-24C 03, SJ-6 05A, SJ-6 05B	SY-24C 01, SY-24C 02, SY-24C 03, SJ-6 05A, SJ-6 05B	The five satellites conducted RPOs from March 2024 through the end of April 2024, at times separated by less than 1 km, and undertaking two simultaneous proximity events. In September 2024, SY-24C 02 and SY-24C 03 conducted three separate approaches, with two being multi-day actions. In December 2024, SY-24C 03 and SJ6 5A came within "tens of meters" of each other five different times; this was a multi-day event.
Jun. 2025 - Jan. 2026	SJ-25	SJ-21	The two satellites were conducting RPOs with each other June 13-14 and at times were unable to be distinguished from each other. They again appeared to come close enough to dock on June 30. They got so close to each other between July 2 and July 6 that they were thought to have docked. The SJ-21 and SJ-25 remained docked until November 2025, at which point they separated but continued to conduct RPOs until mid-January 2026.
Sept. 2025	USA 336	SY-12 02	SY-12 02 conducted an RPO, getting within 60 km of USA 336, possibly in an attempt to image it.

ENDNOTES

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